
Where Does Platonic Convergence Live? A Linear Readout Gains Where the Galaxy Embedding Bends Toward the Label and Loses Where It Bends Away

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that con-
4 vergence is concentrated where the embedding manifold is flat. A pre-registered
5 raw-decoder mean-curvature statistic is negatively associated with cross-survey
6 agreement, but density removes most of the signal and the one residual case disap-
7 pears once its radial component is removed. A second-order expansion of a linear
8 map restricted to the manifold explains why: a readout sees $\langle w_N, \Pi \rangle$, the second
9 fundamental form contracted with its normal component; an alignment's second-
10 order error is a connection mismatch plus a relative extrinsic term; mean curvature
11 is only the metric trace of Π and does not in general determine either. On 86,471
12 labelled galaxies a linear readout gains accuracy where the manifold bends toward
13 the label's own curvature and loses it where the manifold bends away. In a coun-
14 terfactual local second-order surrogate of the readout, the manifold's bending in
15 the readout's normal direction raises local R^2 at 79–99% of anchors, its sign re-
16 versal lowers it at 97–100%, and a random orientation of matched strength shows
17 no such asymmetry and is typically harmful under either sign, on five encoders;
18 across anchors, the mismatch between the label Hessian and the probe-facing Hes-
19 sian is negative wherever the label has local structure and survives cross-fitting and
20 a change of encoder. Mean curvature of a single representation does not explain
21 weak Platonic convergence; the obstruction is relative and, for the fitted global
22 linear alignment, present at first order.

1 Introduction

24 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
25 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on galaxies
26 imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC), which share only 0.3 to 2.3
27 percent of nearest neighbours across surveys under a family of vision models. We ask where on the
28 manifold that agreement is concentrated: if two models disagree most where the embedding surface
29 bends sharply, local mean curvature should associate negatively with local cross-survey agreement.

30 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
31 trained decoder, validated against analytic truth; (ii) a density-controlled test of the curvature–
32 alignment prediction, which finds no density-free effect; (iii) the residual expansion of a linear map
33 on a submanifold; and (iv) known-surface, cross-encoder and interventional tests on the readout side:
34 a linear readout gains local accuracy where the manifold bends toward the label's curvature and loses
35 it where the manifold bends away, at a large majority of anchors on five encoders.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019] (Platonic Universe release [Dupaphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey galaxies embedded by a supervised ImageNet-21k ViT-B/16 (“ViT-B” below; not DINOv3) with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. We unit-normalize embeddings (data on S^{767}).

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; we evaluate curvature on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$ (un-normalized trace; some authors divide by d), by automatic differentiation (cf. [Lee and Park, 2023, Acosta et al., 2023]), at points where DF has full column rank d (likewise $D(F/\|F\|)$ after projection). On the unit sphere the Euclidean mean-curvature vector of every d -submanifold has the universal radial component $-d\hat{x}$, which says nothing about shape; differentiating $F/\|F\|$ instead splits H exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}), and H^S carries the non-radial mean-curvature component. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the projected decoder.

Validation. Against closed-form curvature, at $D = 768$ an in-sphere surface $G(z) = Q \text{normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h four Gaussian bumps, Q a rotation) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H^S\|$, direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

3 Curvature and cross-survey convergence

We correlate the per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings with $\|H_{\text{raw}}\|$ at $d \in \{20, 25, 32\}$. The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is $-0.112, -0.128$ and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}, < 10^{-5}, 0.009$). But k -NN statistics are density-sensitive [Radovanović et al., 2010]: local density correlates with agreement at -0.21 at every d and with $\|H_{\text{raw}}\|$ at $+0.43, +0.32$ and $+0.01$, so residualizing on density leaves $-0.024, -0.066$ and -0.022 . Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; CKA [Kornblith et al., 2019] agrees), and subtracting the radial projection of H_{raw} (a projection of H_{raw} , not a recomputation on $F/\|F\|$) collapses it to -0.023 .

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$ [Lee, 2018], so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

For $M \subset S^{D-1}$, $\Pi = \Pi^S - g\hat{x}$ splits off the sphere’s radial direction, so $\text{Hess}_M(w \cdot x) = \langle w_S, \Pi^S \rangle - (w\hat{x})g(2)$, with w_S the part of w_N orthogonal to $\hat{x}$: a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 , with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$ (all tensor norms and cosines use $\langle A, B \rangle_g = g^{ia} g^{jb} A_{ij} B_{ab}$),

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_g^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A\Pi_F](u, u) +$

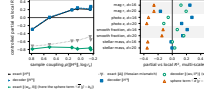


Figure 1: (a) Known surface, linear label, exact geometry: three-control partial vs local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder's $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_S, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control. * not significant at 0.05.

81 $O(|u|^3)$, with Γ the induced connections: the second-order residual contains a connection mismatch
82 and a *relative* extrinsic term, whose target-normal component is $\Pi_G - P_N^G A \Pi_F$.

83 So $\|H\|$, the metric trace's norm, need not correlate with either term; and since the second-order
84 term scales as s^4 and fixed- k neighbourhoods let density set s , patch-scored curvature effects are
85 density-sensitive.

86 5 Probe-facing curvature: bending toward the label helps, bending away 87 hurts

88 Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$
89 from embedding to label; local R^2 inside each of 512 anchors' 2,048-neighbourhoods; rank-partial
90 Spearman against local R^2 with Freedman-Lane p -values [Freedman and Lane, 1983], controlling
91 for log patch radius, local label variance and count and, stronger, log radius at five scales.

92 **Known surface.** Resampling the validation surface with weight $\|H^S\|^\gamma$ (Figure 1a) moves the
93 $\|H^S\|$ partial from -0.29 to $+0.26$ while $\|\langle w_N, \Pi \rangle\|$ holds at -0.74 to -0.80 , there dominated by
94 the sphere term of Eq. (2).

95 **Galaxies.** Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing
96 Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares
97 quadratic of the label on the tangent-projected coordinates $u = g^{-1} J^\top (x - x_0)$ of its 2,048 neigh-
98 bours (decoder-based Monge coordinates, so the finite-patch coefficient estimates the covariant Hes-
99 sian). Three regularities follow (Table 1; Appendix A); the sphere term is negative on every label
100 and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B)
101 it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and
102 stellar mass. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar
103 mass is null throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting,
104 Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). The shape term
105 alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for red-
106 shift): with $K = \langle w_N, \Pi \rangle$, $\|\Delta\|_g^2 = \|\text{Hess}_M y\|_g^2 - 2\langle \text{Hess}_M y, K \rangle_g + \|K\|_g^2$, so the cross term
107 decides for infinitesimal scaling from zero and $\|K\|^2$ is the finite-magnitude overshoot penalty. The
108 same regression on the readout $w \cdot x$ returns the point cloud's own $\langle w_N, \Pi \rangle$ (median cosine with the
109 decoder's 0.55–0.65).

110 **Bending toward helps, bending away hurts.** Within the quadratic surrogate of Eq. (3), the
111 Hessian-mismatch term is lower than at $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle_g > \|K\|_g^2$: the manifold must
112 bend toward the label's own curvature, and not too far. (Liu et al. [2023] derive the optimal normal

weight of least squares on a hypersurface as a uniqueness and stability condition; manifold capacity [Chung et al., 2016, Slatton et al., 2026], flattening [Psenka et al., 2024], latent-curvature studies [Kaufman and Azencot, 2023] and local-regression bias [Cheng and Wu, 2013] use dimension, radius or magnitude; none conditions on the sign of $\langle \text{Hess}_M y, K \rangle$. The partials compare anchors, and no anchor is flat, so we test the condition with a counterfactual local second-order surrogate of the readout. Writing $K = K_S - (w \cdot \hat{x})g$ with $K_S = \langle w_S, \Pi^S \rangle$ the in-sphere shape term, each anchor’s 2,048 neighbours are scored by $\hat{y}_t(u) = c + a^\top u + \frac{1}{2}K_{\text{sph}}(u, u) + \frac{t}{2}K_S(u, u)$ with a the readout’s tangent gradient, K_S from the decoder and c refit: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction (“bending toward the label” means $\langle R, K_S \rangle > 0$ for the remaining label curvature $R = \text{Hess}_M y - K_{\text{sph}}$), $t = -1$ the sign reversal of that probe-facing shape quadratic, $t = 0$ shape-flat. In the Gaussian model $t = 1$ beats shape-flat iff $2\langle R, K_S \rangle_g > \|K_S\|_g^2$; on the actual neighbours, with r_0 the centred shape-flat residual and q_c the centred quadratic $\frac{1}{2}K_S(u_i, u_i)$, the exact criterion is $2\langle r_0, q_c \rangle > \|q_c\|^2$ with optimum $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ (Appendix E). On five encoders and all four labels, $t = 1$ beats shape-flat at 79–99% of anchors and $t = -1$ is worse at 97–100%, while a random orientation with the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign ($t = 1$ beats shape-flat at 22–36% of anchors, $t^* \approx 0$; Appendix E). The probe-selected shape quadratic is locally error-reducing; under the second-order model that is the signature of bending toward the remaining label curvature. This within-anchor question differs from Table 1’s cross-anchor one (does variation in mismatch explain variation in local R^2) and can hold where that has no signal, as for stellar mass. t^* sits at 1.5 to 3.6 throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial second-order term.

Across encoders. The quadratic-chart estimator’s mean-bending trace at the same anchors of five encoders (the readout set’s supervised ViT-B, DINOv3 ViT-B/16, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$) has density-controlled associations with the magnitude probe’s local R^2 of both signs (−0.24 to +0.27), as Eq. (1) allows. Per encoder (Appendix D) the decoder test is stable where the expansion predicts: the mismatch partial is negative for magnitude and redshift on all five encoders (−0.32 to −0.61; cross-fitted −0.28 to −0.62) and the alignment positive for magnitude on all five (+0.13 to +0.53) and for redshift on four; the shape term’s sign varies with encoder as well as label; for the mismatch and alignment hypotheses stellar mass stays null but for one mismatch cell (−0.12).

6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold, seen in the readout’s normal direction, bends toward the label’s curvature and loses it where that bending is reversed: the gain at a large majority of anchors, the loss at nearly every anchor, on five encoders. The residual expansion names the quantity, $\langle w_N, \Pi \rangle$ against $\text{Hess}_M y$, and a decoder’s pointwise second fundamental form measures it at production scale, validated against analytic truth for H^S and in substantial agreement with the data. For a linear alignment map [Bangachev et al., 2026] the relevant second-order extrinsic normal term is $\Pi_G - P_N^G A \Pi_F$, not one embedding’s mean curvature; the full residual also carries the connection mismatch; our pilot (Appendix C) finds a 44% first-order tangent-frame residual, so its relative- Π column is only an extrinsic diagnostic, and no estimator-stable second-order association. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10–0.17); its norm is reliably ranked (split-half ρ 0.91–0.94) but its direction only moderately (tensor cosine 0.19–0.35). On a synthetic surface with known curvature the mismatch partial reproduces but the alignment partial has no stable sign; Appendix E carries the alignment claim. Mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

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Appendix A: decoder ablations

Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach variance explained 0.952). Every significant sign of the main table is retained.

label	quantity	seed 0	seed 1	seed 2	400^3
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	-0.26	-0.26	-0.27	-0.26
	mismatch	-0.39	-0.35	-0.34	-0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	-0.06*	-0.09*	-0.09	-0.06*
	sphere	-0.33	-0.35	-0.34	-0.35
	mismatch	-0.45	-0.43	-0.42	-0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	-0.13	-0.15	-0.14	-0.11
	mismatch	-0.09*	-0.05*	-0.08*	-0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	-0.06*	+0.05*	+0.09
	sphere	-0.35	-0.35	-0.35	-0.35
	mismatch	-0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	-0.02*	+0.03*	-0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

Appendix B: sensitivity of the mismatch and sphere columns

Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	-0.38	-0.34	+0.33	+0.31	+0.35	-0.21 (-0.26)
mag_r	20	-0.43	-0.40	+0.39	+0.37	+0.24	-0.22 (-0.26)
photo_z	16	-0.42	-0.36	+0.23	+0.20	+0.32	+0.05* (-0.33)
photo_z	20	-0.42	-0.42	+0.22	+0.22	+0.20	+0.03* (-0.36)
smooth_fraction	16	-0.10	-0.17	+0.17	+0.19	+0.33	-0.12 (-0.13)
smooth_fraction	20	-0.15	-0.21	+0.05*	+0.03*	+0.23	-0.13 (-0.15)
stellar_mass	16	+0.01*	-0.04*	-0.03*	+0.06*	+0.31	-0.02* (-0.35)
stellar_mass	20	-0.06*	-0.07*	+0.01*	-0.00*	+0.19	-0.02* (-0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

Appendix C: relative second fundamental form, pilot

Two sphere-projected decoders ($\text{HSC} = F$, $\text{Legacy} = G$; Phase 7 protocol), a global ridge map A from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ; median first-order obstruction $\|AJ_F - J_GL\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of each decoder's $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$ (because first-order

matching $AJ_F = J_G$ fails at the 44% level, the second derivative of the alignment residual is not coordinate invariant; in the decoder chart its target-normal quadratic coefficient additionally contains $-P_N^G AJ_F \Gamma^F$, so relative II is reported only as an extrinsic diagnostic), the same with L fitted on 256 neighbours' latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F 's tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	II _{rel}	II _{rel} (local L)	II _{rel} (data)
$d = 20$, seed 0	0.64	0.44	-0.15	+0.04*	-0.18	+0.04*	-0.18	+0.13
$d = 25$, seed 0	0.64	0.43	-0.10	+0.06	-0.16	+0.05	-0.23	+0.13
$d = 20$, seed 1	0.64	0.44	-0.11	+0.05	-0.19	+0.04	-0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

234

235 Appendix D: the same test on four further encoders

236 Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders
 237 from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024,
 238 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed
 239 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$,
 240 multi-scale density control. The first row per label is the probe's global out-of-sample R^2 ; Table 6
 241 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per
 encoder.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	-0.11	+0.16	+0.16	-0.21
	sphere	-0.26	-0.23	-0.40	-0.24	-0.08*
	mismatch	-0.39	-0.57	-0.34	-0.32	-0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	-0.06*	-0.05*	-0.17	+0.29	-0.16
	sphere	-0.33	-0.21	-0.29	-0.46	-0.44
	mismatch	-0.45	-0.61	-0.45	-0.48	-0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	-0.08*	+0.20	+0.24	+0.06*
	sphere	-0.13	-0.06*	+0.05*	-0.15	-0.11
	mismatch	-0.09*	-0.33	-0.25	-0.20	-0.01*
	alignment	+0.19	-0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	-0.15	-0.10	-0.11	-0.13
	sphere	-0.35	-0.26	-0.38	-0.40	-0.38
	mismatch	-0.04*	+0.03*	-0.12	-0.06*	-0.02*
	alignment	+0.01*	-0.05*	-0.01*	+0.07*	-0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

242

243 Appendix E: intervening on the readout's normal component

244 The partials of Table 1 compare anchors with one another and cannot say whether reversing the probe-
 245 facing shape term hurts relative to removing it, since no anchor is flat. Here we score a counterfactual
 246 local second-order surrogate of the readout at a fixed manifold. At each anchor the fitted ridge weight
 247 is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$; we score the anchor's
 248 2,048 neighbours by the data-side tangent-plus-sphere readout $(w_T + w_{\text{rad}}) \cdot x$ plus tq with the
 249 local intercept refit, where $q = \frac{1}{2} \langle w_S, \Pi^S \rangle(u, u)$ is the decoder's in-sphere second-order term in the
 250 anchor's chart coordinates: $t = 1$ is the manifold's bending as seen in the readout's normal direction,
 251 $t = -1$ its sign reversal, $t = 0$ shape-flat (the sphere term $-(w \cdot \hat{x})g$ is kept in the base). Because

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

252 the intercept is refit for every t , the local sum of squares is exactly $\text{SSE}(t) = \|r_0 - tq_c\|^2$ with r_0
 253 the centred residual of the shape-flat predictor and $q_c = q - \bar{q}\mathbf{1}$ the centred quadratic on the anchor's
 254 neighbours; hence $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ and $t = 1$ beats shape-flat iff $2\langle r_0, q_c \rangle > \|q_c\|^2$. This is
 255 the finite-sample counterpart of the tensor-level condition $2\langle R, K_S \rangle_g > \|K_S\|_g^2$ of Section 5, not
 256 the same quantity: the tensor version needs the estimated label Hessian, the empirical one does not.
 257 Columns: fraction of anchors where $t = 1$ beats shape-flat (help) and where $t = -1$ is worse than
 258 shape-flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ; then the same for a random
 259 in-sphere normal direction v whose quadratic is rescaled to the same centred amplitude on the actual
 260 neighbours, $\|q_{v,c}\| = \|q_c\|$ (matching $\|v\|$ to $\|w_S\|$ instead leaves the contracted tensor at a fraction
 261 of the fitted one, since Π^S spans at most $d(d+1)/2$ of the ~ 750 normal directions; matching
 262 $\|\langle v, \Pi^S \rangle\|_g$ gives the same picture; Supplement 12). The anchors' neighbourhoods overlap (each
 263 point lies in about twelve of them), so the fractions are not counts of independent trials; a maximal
 264 set of anchors whose pairwise overlap is at most 5% of k has 19–21 members per run, on which $t = 1$
 265 beats shape-flat at 81–100% (one-sided sign test $p \leq 4 \times 10^{-3}$ in every cell) and $t = -1$ is worse
 266 at 95–100% ($p \leq 3 \times 10^{-5}$).

run	label	decoder in-sphere shape term					random orientation, matched $\ q_c\ $				
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	-0.051	2.3	0.26	0.80	-0.009	-0.009	
	photo_z	0.99	0.99	+0.028	-0.038	3.0	0.33	0.68	-0.005	-0.006	
	smooth_fraction	0.90	1.00	+0.021	-0.040	1.7	0.24	0.78	-0.008	-0.010	
	stellar_mass	0.96	0.99	+0.026	-0.040	2.6	0.29	0.70	-0.006	-0.006	
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	-0.049	2.6	0.24	0.78	-0.007	-0.008	
	photo_z	0.99	1.00	+0.029	-0.038	3.6	0.36	0.71	-0.004	-0.005	
	smooth_fraction	0.90	1.00	+0.020	-0.035	1.9	0.22	0.77	-0.007	-0.008	
	stellar_mass	0.97	1.00	+0.027	-0.038	3.1	0.32	0.72	-0.005	-0.005	
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	-0.031	2.0	0.29	0.71	-0.007	-0.006	
	photo_z	0.96	0.99	+0.020	-0.030	2.6	0.29	0.71	-0.006	-0.004	
	smooth_fraction	0.93	1.00	+0.024	-0.055	1.5	0.22	0.78	-0.014	-0.013	
	stellar_mass	0.95	1.00	+0.016	-0.028	2.1	0.26	0.74	-0.005	-0.005	
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	-0.031	2.2	0.30	0.68	-0.006	-0.005	
	photo_z	0.99	1.00	+0.019	-0.026	3.4	0.36	0.64	-0.003	-0.003	
	smooth_fraction	0.92	0.99	+0.014	-0.027	1.7	0.25	0.69	-0.006	-0.005	
	stellar_mass	0.97	1.00	+0.020	-0.028	2.8	0.35	0.68	-0.004	-0.005	
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	-0.031	2.0	0.26	0.73	-0.007	-0.007	
	photo_z	0.99	1.00	+0.018	-0.026	2.9	0.28	0.64	-0.004	-0.003	
	smooth_fraction	0.88	0.98	+0.016	-0.036	1.6	0.23	0.75	-0.009	-0.008	
	stellar_mass	0.99	1.00	+0.022	-0.032	3.1	0.29	0.65	-0.005	-0.003	
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	-0.034	2.6	0.32	0.74	-0.005	-0.006	
	photo_z	0.97	0.99	+0.018	-0.025	3.2	0.34	0.67	-0.004	-0.003	
	smooth_fraction	0.79	0.97	+0.013	-0.029	1.6	0.25	0.77	-0.006	-0.008	
	stellar_mass	0.98	1.00	+0.022	-0.032	2.9	0.30	0.67	-0.004	-0.004	

Table 7: Counterfactual scaling of the in-sphere shape term in a local second-order surrogate of the readout, 512 anchors per run. The manifold’s bending as seen in the readout’s normal direction helps at a large majority of anchors, its sign reversal hurts at nearly every anchor, a random orientation of the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign; $t^* > 1$ throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial term.